

Exercise 2: SQL Aggregates AND Operations

1. SELECT DISTINCT Department
FROM students;

Output

Department
IT
HR
Finance

2. SELECT department,
AVG (age) AS avg-age
FROM students
Group BY department;

Output

department	avg-age
IT	21
HR	22
Finance	23

3. SELECT department,
COUNT (department) AS student-count
FROM students
WHERE student-count > 1

GROUP BY department;

Output

department	student_count
IT	2
HR	2

```
4. SELECT student_id,  
           name,  
           age,  
           department  
FROM students  
WHERE age BETWEEN 21 AND 23
```

Output

student_id	name	age	department
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	22	HR

```
5. SELECT student_id,  
           name,  
           age,  
           department  
FROM students
```


WHERE department IT OR HR AND age > 21;

Output

student_id	name	age	department
2	Bob	22	HR
4	Diana	23	finance
5	Eve	22	HR

6. SELECT department,
SUM(credits) AS total_credits
FROM courses
WHERE total_credits > 5
GROUP BY department;

department	total_credits
IT	11

7. SELECT course_id,
course_name,
department,
credits
FROM courses
WHERE credits < 4

course_id	course_name	department	credits
101	SQL basics	IT	3

104	Excel	Finance	2
105	Statistics	HR	3

8. SELECT course_id,
course_name,
credits
FROM courses
ORDER BY credits DESC
LIMIT 3;

Output

course_id	course_name	credits
102	Python	4
103	Data Science	4
101	SQL Basics	3
105	Statistics	3

9. a) SELECT MAX(grade) AS max_grade
FROM enrollments;

Output

max_grade
90

b) SELECT MIN(grade) AS min_grade

FROM enrollments

Output

min_grade
78

9) SELECT AVG(grade) AS avg_grade
FROM enrollments;

Output

avg_grade
85

10. SELECT COUNT(enrollment_id) AS enrollment_count,
course_id
FROM enrollments
GROUP BY course_id;

Output

course_id	enrollment_count
101	1
102	1
103	1
104	4
105	1

11. SELECT department,

```

SUM (salary) AS total_salary,
SUM (bonus) AS total_bonus
FROM salaries
GROUP BY department

```

Output

department	total_salary	total_bonus
IT	122 000	10 500
HR	109 000	7 500
Finance	70 000	6 000

```

12 SELECT department,
    AVG(salary) AS avg_salary
FROM salaries
WHERE avg_salary > 55000
GROUP BY department;

```

Output

department	avg_salary
IT	61 000
Finance	70 000

reworked

13. SELECT employee_id,
 name,
 salary,
 bonus,
 (salary + bonus) AS total_compensation
 FROM salaries
 WHERE total_compensation > 60 000 ;

Output

employee_id	name	salary	bonus	total_compensation
1	Tom	60 000	5 000	65 000
3	Spike	70 000	6 000	76 000
4	Tyke	62 000	5 500	67 500

14. SELECT department,
 SUM(budget) AS total_budget,
 AVG(budget) AS avg_budget
 FROM projects
 WHERE ~~budget > 70 000~~ avg_budget > 70 000

department	total_budget	avg_budget
IT	270 000	135 000
Finance	80 000	80 000

15. SELECT project_id,
 project_name,

department,
budget
FROM projects
WHERE budget BETWEEN 50 000 AND 120 000
AND department NOT IN ('Marketing')

Project-id	project-name	department	budget
1	AI App	IT	120 000
2	Payroll System	Finance	80 000
3	Dashboard	IT	150 000
5	HR Portal	HR	50 000